

Nested For Loops

```
for i in range(3):  
    for j in range(3):  
        print(i, j)
```

Atharva Madan Jadhav

CS3 – C35

PRN - 202501040350

Subject - EDS

Topic – Assignment-1 (Nested For Loops with 1 example)

<https://youtu.be/eyOYKA3oBVE> -YT link

What is a For Loop?

Definition

A **for loop** repeats a block of code for every item in a sequence (list, range, string, etc.).

Basic Syntax

```
for variable in sequence:  
    # code block (indented)  
    statement
```

► Iterates over a sequence

Runs once per element

► Indentation is mandatory

Defines the loop body

► range(n) generates numbers

0 to n-1 by default

Nested For Loop — Core Concept

A **nested for loop** is a for loop placed **inside another for loop**. For every single iteration of the outer loop, the inner loop runs completely from start to finish.

Syntax

```
for i in outer_sequence:
    for j in inner_sequence:
        # innermost code
```

Outer Loop

Controls the rows / major iterations. Variable *i* changes each outer cycle.

Inner Loop

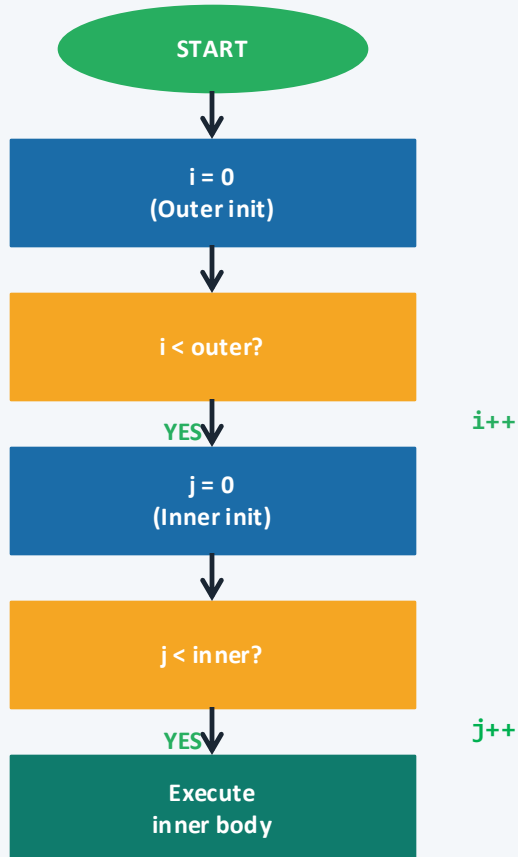
Runs completely for EVERY single step of the outer loop. Variable *j* resets each time.

Inner Body

Executes (outer iterations × inner iterations) times in total.

 Total Executions = **outer iterations** × **inner iterations** e.g. `range(3) × range(4)` = 12 total executions

Execution Flow Diagram



1

Outer Loop Starts

i initialises to 0. The outer condition ($i < \text{outer limit}$) is checked.

2

Inner Loop Triggers

For each valid i, j resets to 0 and the inner loop begins checking its condition.

3

Inner Body Executes

The innermost statements run. j increments after each iteration.

4

Inner Loop Exhausted

When j condition fails, control returns to the outer loop; i increments.

5

Outer Loop Exhausted

When i condition fails, the nested loop ends completely.

Simple Example: Multiplication Table



Task: Print a 3×3 multiplication table using a nested for loop.

Python Code

```
# Multiplication Table (3x3)
for i in range(1, 4):
    for j in range(1, 4):
        print(i, "x", j, "=", i*j)
```

Output

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
```

Outer loop (i)

i runs: 1 → 2 → 3
(3 iterations total)

Inner loop (j)

j runs 1→2→3 for EACH i
(3 × 3 = 9 prints)

range(1,4)

Generates 1, 2, 3
(stops before 4)

Problem Analysis — Iteration Trace (How Python Executes It)

Step	Outer i	Inner j	$i \times j$	Output Printed
1	1	1	1	$1 \times 1 = 1$
2	1	2	2	$1 \times 2 = 2$
3	1	3	3	$1 \times 3 = 3$
4	2	1	2	$2 \times 1 = 2$
5	2	2	4	$2 \times 2 = 4$
6	2	3	6	$2 \times 3 = 6$
7	3	1	3	$3 \times 1 = 3$
8	3	2	6	$3 \times 2 = 6$
9	3	3	9	$3 \times 3 = 9$



i = 1 cycle



i = 2 cycle



i = 3 cycle

Complex Problem Solving — Pyramid Star Pattern

✿ Printing a pyramid of stars with n rows — a pattern problem that requires two loop variables working together.

Python Code

```
n = 5
for i in range(1, n+1):
    for j in range(i):
        print("*", end="")
    ) # end= avoids newline
    print() # move to next line
```

Output

```
*
**
***
****
*****
```

🔑 Inner range(i)

Each row prints exactly i stars — the inner loop count equals the row number.

🔄 end=""

Prevents print() from adding a newline, keeping stars on the same row.

🚩 Outer print()

Forces a new line after each full row of stars is printed.

Real-World Applications — Where Nested Loops Are Used



Image Processing

Outer loop → rows of pixels
Inner loop → columns of pixels
Each pixel's color is modified.



Matrix Operations

Outer loop → matrix rows
Inner loop → matrix columns
Used in data science & ML.



Game Grids

Outer loop → grid rows
Inner loop → grid columns
Draw boards, tile maps, etc.



Report Generation

Outer loop → each student
Inner loop → each subject
Print grade sheets, tables.



Search Algorithms

Outer loop → one element
Inner loop → compare with rest
Bubble sort, selection sort.



Pattern Printing

Outer loop → row number
Inner loop → symbol count
Star, number, diamond patterns.

Common Mistakes & Best Practices

Variable Name Clash

Wrong

Using same variable for both loops:

```
for i in range(3):  
    for i in range(3):
```

Correct

Use distinct variables:

```
for i in range(3):  
    for j in range(3):
```

Indentation Error

Wrong

```
for i in range(3):  
for j in range(3): # not indented!
```

Correct

```
for i in range(3):  
    for j in range(3): # indented ✓
```

Performance Concern

Wrong

```
# n=1000 → 1,000,000 iterations  
# Very slow for large n!
```

Correct

```
# Minimize nesting depth  
# Use break/continue wisely
```

Nested For Loops — Summary

01 A nested for loop is a loop inside another loop — the inner loop completes fully for each outer iteration.

02 Total iterations = outer × inner. Understanding this prevents performance issues.

03 Key syntax rules: distinct variable names (i, j), correct indentation, proper use of range().

04 Used in multiplication tables, star patterns, image processing, matrix operations, and sorting algorithms.

05 Always trace through iterations step by step to verify correctness before running the code.